

Hyperspectral Image Classification using Hybrid Approach

*Note: This is the report of our paper, which we are currently working on, this is not the final paper.

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Abstract—As the novelty of this work, we present a hybrid deep learning model which is a combination of GNNs and ViTs for hyperspectral image classification. Our approach takes the best of both worlds - dealing with long-range dependencies using transformers and local spatial relationships via graph-based learning. We propose an unsupervised pretraining approach based on masked image modeling (MIM) to learn rich spectral-spatial representations with the fine-tuning for classification. In the experiments on the challenging Houston 2018 dataset, our method achieves remarkable accuracy of 98.52%, showing significantly better performance than traditional methods. The proposed network is suitable for coping with the high dimension and limited labeled samples in hyperspectral data. Our model not only achieves state-of-the-art accuracy for hyperspectral image classification, but also demonstrates an architecture that can also potentially be generalized to other remote sensing applications with complex spectral-spatial correlation to be captured efficiently.

Index Terms—Graph Neural Networks, Vision Transformers, Hyperspectral Image, Machine Learning, Cross-attention, patch embedding

I. INTRODUCTION

Advanced remote sensing systems have revolutionised the monitoring and study of the Earth, and hyperspectral imaging (HSI) has become an exceptionally powerful technology for mapping and detailed analysis of the Earth's surface. Unlike conventional multispectral sensors that record a few broad bands, hyperspectral sensors sample the reflectance spectrum with hundreds of contiguous, narrow bands, exposing the subtle material properties unobservable to conventional sensors. This spectral diversity allows applications such as precision agriculture and mineral exploration, urban planning and ecological assessment with an unprecedented level of detail. However, the very characteristic that makes hyperspectral data valuable—its

high dimensionality—also creates significant analytical challenges that conventional classification approaches struggle to overcome.

The key difficulty of hyperspectral image classification lies in the intricate interaction between the spatial and spectral dimensions, and the well-known "Hughes phenomenon" that classification accuracy degrades as dimensionality exceeds a certain threshold compared to the number of available training samples. The traditional ways are indirect sometimes to extract the complicated spectral-spatial information or the amount of labeled data are large and hard to gain by human manual annotation in the remote sensing. Recent deep learning approaches have shown promise but typically emphasize either spectral or spatial characteristics without effectively integrating both domains in a unified framework that can leverage limited training data.

This paper attempts to construct a new kind of hybrid deep learning architecture effectively combining GNN and ViT to well model the local and global information within hyperspectral imagery. Our aims in this work are: (1) proposing a self-supervised pre-training approach with masked image modeling to extract effective representations from unlabelled data; (2) devising an end-to-end GNN-ViT framework to capture intricate spectral-spatial patterns; and (3) validating our approach on the challenging Houston 2018 dataset to demonstrate its effectiveness for urban land cover classification with limited training samples.

The implications of this study go beyond enhancing the performance of classification accuracy measures. Our work is expected to improve the availability and practicality of advanced remote sensing techniques for real applications by combating inherent limitations in hyperspectral image analysis. The capability to mine reliable knowledge from hy-

perspectival data using small portions of known samples has a broad range of applications in environmental monitoring, disaster relief and sustainable resource management, which often require real-time analysis but have limited reference data. Furthermore, our hybrid architecture and self-supervised learning approach provide a basis for further exploration in multi-sensor data fusion and transfer learning across diverse remote sensing domains.

II. MODEL ARCHITECTURE

This section entails a detailed description of our approach. The model first uses a Masked Image Modelling (MIM) pretraining approach before fine-tuning on the classification task. The architecture leverages both the spatial relationships captured by transformers and the graph-based relationships captured by GNNs. It contains an explanation of all the components that we have used to surpass previous approaches:

A. Dataset Handling

We have created a custom class that enables us to handle the data efficiently. This specialized class handles the complex Houston 2018 hyperspectral dataset with its 48 spectral bands and 20 land cover classes. We utilized h5py to handle newer and scipy.io to handle older formats of the data. The data was then normalized to a range of [0,1] and then the dataset's original shape is transformed from spatial-first ($H \times W \times C$) to channel-first format ($C \times H \times W$), aligning with the expected input structure. After that, it was separated into 8×8 patches with the proper padding around labelled pixels. By maintaining the spatial context surrounding each target pixel, this method enables the model to use contextual information to classify data more accurately. The custom_collate function helped pass tensors of the same size into the model.

Parameter	Description	Value Used
data_path	Path to the data file containing spectral bands	Path from config: \Houston18.mat
label_path	Path to the file containing ground truth labels	Path from config: \Houston18_gt.mat
transform	Optional transformations to apply to the data	Default: None
patch_size	Size of patches to extract from the image	4 (from config)

TABLE I

DATA CONFIGURATION PARAMETERS (SIMPLIFIED PATHS)

custom_collate

Parameter	Description	Value Used
Batch	List of sample tuples (data, target)	Dynamic from DataLoader

TABLE II

BATCH CONFIGURATION PARAMETER

(NOTE: This part will be edited once before final paper submission since it only contains the description about

the houston2018 dataset. CONTENT TO BE ADDED: Information about other datasets that we have used)

B. Loss Function

We have used Weighted Binary Cross Entropy Loss (WBCELoss) for our loss function since it addresses one of the most important problems in remote sensing data classification that is class imbalance. The function transforms the integer numbers into one-hot vectors so that each class can be trained using its own method. A factor is then added to the formula (default value is set to 0.5) to balance the representation of the majority and minority classes. Positive examples get weight $(1 - \text{ratio})$ and negative ones get weight ratio. In this way, we eliminate the difficulty of different class frequencies. It helps the model by making gradient signals more stable when the number of samples in each class is very different. Because of this, minority groups are shown more clearly in the data so that the model does not favour the main classes.

Parameter	Description	Value Used
ratio	Weight ratio for positive vs negative samples	0.5 (from config)

TABLE III

RATIO PARAMETER CONFIGURATION

C. Vision Transformer Components

1) *Patch Embedding* : This module serves as an entry point for the HSI dataset into the main transformer architecture. It converts raw pixel values into high dimensional token representation. Patch embedding reshapes the input tensor to a single vector representation that is prepared for convolution operations. The convolution layer has kernel size that matches the full spectral depth of the dataset and the desired patch size. The stride is chosen in a way to avoid overlapping patches. This function reduces the spatial dimensions of the data as well as projects the features into a higher dimensional embedding space. It carefully reshapes the tensor to the required input dimension (batch_size, num_patches, embedding_dimension) while preserving the spatial arrangement information before applying self-attention mechanisms. Thus, it becomes possible for the transformer to use sequences and the attention mechanism to learn how patches and spectral bands relate to one another when using the hyperspectral data.

Parameter	Description	Value Used
img_size	Size of the input image	8 (2x patch_size)
patch_size	Size of each patch	4
in_channels	Number of input channels	48 (spectral bands)
embed_dim	Dimension of the embedding space	96

TABLE IV

MODEL CONFIGURATION PARAMETERS

2) *MLP Head* : The MLP Head serves a purpose of acting as the feed-forward network for each transformer block which plays a crucial role in the model's representation power. The module consists of two linear projection layers with GELU

activation function in between which is used to separate the layers. Contrary to RELU, GELU exhibits a better non-linearity and is also better for calculating gradients. Depending on the `mlp_ratio`, the first layer changes the input shape from `in_features` to `hidden_features` which is normally four times greater. Because of this expansion, detecting complex patterns is now simpler due to the increase in our space for representation. The second layer is used to bring the expanded data to its normal dimensions which forces information compression. We have applied a dropout layer after each transformer block which is used to introduce controlled noise during training, preventing co-dependency between specific neurons and improving generalization to new data. After cross-attention, the MLP module in the transformer block examines token representations using an extra computation technique that considers different aspects of the data. While MLP describes the complex, non-linear changes in individual token attributes, attention processes explain the relationships between tokens. This combination enhances the model's capacity to recognize complex spectral-spatial patterns in hyperspectral data, hence improving data classification.

Parameter	Description	Value Used
<code>in_features</code>	Number of input features	<code>embed_dim</code> (96)
<code>hidden_features</code>	Number of hidden features	<code>int(embed_dim * mlp_ratio)</code> (384)
<code>out_features</code>	Number of output features	<code>embed_dim</code> (96)
<code>dropout</code>	Dropout rate	0.1

TABLE V
MLP BLOCK CONFIGURATION PARAMETERS

3) *Cross-attention* : Cross Attention uses a complex attention mechanism to enable the model to reason about contexts. Unlike standard self-attention, the upgraded module can perform self-attention or cross-attention according to the data it receives. The module first projects input features into query, key, and value representations using separate linear transformations, allowing different parts of the features to be emphasized in each role. At this point, the model is able to look at various patterns in relationships by redistributing the attention of the heads to several channels of interest. Scaled dot-product attention is applied to find the attentions needed. To obtain attention weights, the similarities between a query and a key are computed, scaled slightly with the square root of the head size and finally normalized by SoftMax. Weights allow us to blend together several value vectors according to how much each token affects them. The module makes it possible for information to pass from the visible regions to the masked regions during pre-training and then from patch tokens to class tokens in the fine-tuning stage, thanks to cross-attention across features. With the multi-head approach, the model is better able to identify complex connections since it processes attention in multiple representational areas at the same time. Change the way information travels through

the network by switching from local to global reasoning, an attention mechanism helps the model make use of relationships among different positions in the hyperspectral image.

Parameter	Description	Value Used
<code>dim</code>	Dimension of input features	<code>embed_dim</code> (96)
<code>num_heads</code>	Number of attention heads	8
<code>qkv_bias</code>	Use bias in QKV projections	True
<code>attn_drop</code>	Dropout rate for attention	0.1
<code>proj_drop</code>	Dropout rate for projections	0.1

TABLE VI
ATTENTION MODULE CONFIGURATION PARAMETERS

4) *Transformer Block* : This module is the core component of the transformer, which combines the feed-forward networks and the attention mechanisms in a residual architecture. This component implements a pre-norm design which applies layer normalization before attention mechanisms and feed-forward networks which helps in stabilizing the training of the model by restricting the feature distributions. The cross-attention mechanism in the first branch of the model takes in input data that has been normalized to establish relationship links between the tokens. The MLP module is then used in the second layer to document feature transformations. Because both of these branches use residual connections ($x + \text{branch_output}$), which enhance gradient flow during backpropagation, deeper networks can be trained. This structure allows the model to skip attention or MLP layers when advantageous, hence giving the network the flexibility to apply transformations selectively. Transformer blocks gradually improve representations since they are stacked one after the other. Earlier blocks capture the basic spatial-spectral patterns while the deeper blocks model more abstract relationships. The sequencing of this system plays a big role since information can be passed between any points in one layer, however wide the gap between them. This allows the model to identify long-range dependencies in HSI data. This global context integration provides a significant advantage over traditional convolutional methods, which only gradually build global comprehension through layers with confined receptive fields.

D. Graph Neural Network Components

1) *GNN Block* : This module implements a Graph Attention Network (GAT) architecture that passes data patches as nodes in a graph structure. It learns and propagates information around the graph using two layered GATConv layers. By using multi-head attention (`num_heads`) and projecting input characteristics from `input_dim` to `hidden_dim`, the first layer effectively expands the feature space and captures a variety of node interaction aspects. Every attention head in the model independently computes attention coefficients to highlight the importance of the nearby nodes. The Exponential Linear Unit (ELU) activation function is used after the first

Parameter	Description	Value Used
dim	Dimension of input features	embed_dim (96)
num_heads	Number of attention heads	8
mlp_ratio	MLP hidden dim to input dim ratio	4.0
qkv_bias	Use bias in QKV projections	True
drop	Dropout rate	0.1
attn_drop	Dropout rate for attention	0.1

TABLE VII
TRANSFORMER BLOCK CONFIGURATION PARAMETERS

layer to introduce non-linearity in the data while maintaining smooth gradients. After compressing the multi-attention head and transforming the features to the output dimension, the second layer combines the several representations to provide a coherent node embedding. By randomly choosing a fraction of the neurons in each layer during training and deactivating them, dropout between the layers helps avoids overfitting. With this method, how spatial relations are modelled is significantly changed since the graph structure shows what nodes are connected rather than treating all in the same way as in transformers since it allows the model to reason about spatial adjacency and similarity. Because graph operations scale with the number of edges rather than quadratically with the number of nodes, the hybrid architecture can now model complex relationships between spatial regions while still being computationally efficient thanks to the addition of this GNN component.

Parameter	Description	Value Used
input_dim	Dimension of input node features	embed_dim (96)
hidden_dim	Dimension of hidden node features	64
output_dim	Dimension of output node features	embed_dim (96)
num_heads	Number of attention heads in GAT	4
dropout	Dropout rate	0.1

TABLE VIII
GRAPH ATTENTION NETWORK (GAT) CONFIGURATION PARAMETERS

E. Main Model Architecture

1) *HybridGNNViT* : The HybridGNNViT acts as the base model for our mixture of transformer and GNN techniques. To begin classification, the input patches are converted using patch embedding into token embeddings. Data from every patch token goes into forming one large token, referred to as the learnable token. Position embeddings help the model retain spatial information that would otherwise be lost in the transformer's operations. The token sequence then passes through multiple transformer blocks that processes the data with cross attention mechanisms, successfully capturing long term dependencies across multiple input sequences. The data is

handled in the GNN module after passing via the transformer block. In the build_graph method, the graph is constructed so that the edges show the spectral link and the nodes show the patch tokens. Learned attention weights are used to direct information movement between patches instead of being directed based on fixed spatial relationships after the GNN processes the graph. This hybrid approach uses the complementary strengths of each model; the transformer excels at modelling global context through its attention mechanisms, while GNN excels at modelling the structured relationship through the graph. This approach has significantly improved the classification accuracy by capturing diverse aspects of the data.

Parameter	Description	Value Used
img_size	Size of input image	8 (2x patch_size)
patch_size	Size of each patch	4
in_channels	Number of input channels	48 (spectral bands)
num_classes	Number of output classes	20
embed_dim	Dimension of embedding space	96
depth	Number of transformer blocks	4
num_heads	Number of attention heads	8
mlp_ratio	Ratio of MLP hidden dim to input dim	4.0
qkv_bias	Use bias in QKV projections	True
drop_rate	Dropout rate	0.1
attn_drop_rate	Attention dropout rate	0.1
gnn_hidden_dim	Dimension of GNN hidden features	64
gnn_heads	Number of attention heads in GNN	4
gnn_dropout	Dropout rate for GNN	0.1

TABLE IX
COMPREHENSIVE MODEL CONFIGURATION PARAMETERS

2) *MaskedSST_MIM* : The MaskedSST_MIM architecture implements a masked image modelling approach that in turn is used for self-training. It adapts the masked autoencoder approach to HSI data, which helps create a powerful pre-training method that requires no labelled data. The approach begins with a random masking of the data (70%) which removes patch tokens based on random noise values. The remaining tokens are then passes to the transformer block which forces the model to capture rich representations that can be used later to reconstruct the data. The decoder consists of a simple transformer that attempts to reconstruct all the patches including the masked data. The patchify method transforms input images into the patch representation needed for reconstruction. This approach makes it harder for the model to learn simple solutions by minimizing the reconstruction error, especially on the masked patches. This forces the model to develop strong dependency understanding as well as understand the inherent structure of the HSI data. The resulting pretrained model provides an excellent starting point for the classification

model by significantly improving convergence speed and final accuracy.

Parameter	Description	Value Used
img_size	Size of input image	8 (2x patch_size)
patch_size	Size of each patch	4
in_channels	Number of input channels	48 (spectral bands)
embed_dim	Dimension of embedding space	96
depth	Number of transformer blocks	4
num_heads	Number of attention heads	8
mlp_ratio	Ratio of MLP hidden dim to input dim	4.0
qkv_bias	Use bias in QKV projections	True
drop_rate	Dropout rate	0.1
attn_drop_rate	Attention dropout rate	0.1
mask_ratio	Ratio of tokens to mask	0.7

TABLE X
TRANSFORMER MODEL CONFIGURATION PARAMETERS

F. Training Pipeline Components

1) *Pretraining Process* : The pre-training process sets up the self-supervised learning phase of the approach by using masked image modelling. It uses unlabelled data, utilizing the structure of HSI data to learn meaningful patterns and representations. During each epoch the input data that is fed to the MaskedSST_MIM model which handles the random masking of patches, encoding of non-masked patches and reconstruction of the masked regions. The compute_loss function calculates the MSE between the actual vs predicted pixel values, but only for the masked data which creates a bottleneck that prevent the model from memorising the data but instead helps in generalizing it instead. The AdamW optimizer updates model parameters with weight decay, helping avoid overfitting. Without labels, the model learns to understand underlying patterns in the structure of the data, correlation between spectral bands, and spatial patterns. The resulting weight provides a much better starting point for the classification model than just using random weights, especially for the transformer model. This self-supervised learning phase helps improve the supervised phase, resulting in better generalization from limited labelled data and also avoid overfitting to the training set.

2) *Fine-tuning Process* : This module is used to transform the self-supervised pre-trained model into a supervised classification model. It introduces important components for effective transfer learning. The function uses both standard cross-entropy and weighted binary cross-entropy loss (WBCELoss), creating a training signal that addresses the issue of class imbalance in the HSI data. The model uses an early stopping principle that prevents overfitting throughout training. Careful

Parameter	Description	Value Used
model	MIM model to pre-train	MaskedSST_MIM instance
dataloader	DataLoader for training data	train_loader
optimizer	Optimizer for updating weights	AdamW with lr=1e-4, weight_decay=1e-4
device	Device to run on (CPU/GPU)	CUDA if available, else CPU
num_epochs	Number of pretraining epochs	20

TABLE XI
PRETRAINING CONFIGURATION PARAMETERS

monitoring of the model performance based on validation metrics, avoids performance degradation during the training of deeper networks and ensures optimal generalization. The model is updated from prior reconstruction goals to be able to classify different examples, using the previously used features as input. The classifier model is significantly improved by using the MaskedSST_MIM weights. The lower layers continue to collect general features, while the upper layers conform to the structure of the data. In contrast to building the classification model from scratch, using transfer learning improves the final accuracy and speeds up the learning process. A classifier based on this method provides high-performance using both supervised and self-supervised learning.

Parameter	Description	Value Used
model	Model to finetune	HybridGNNViT instance
train_loader	DataLoader for training data	train_loader
val_loader	DataLoader for validation data	val_loader
test_loader	DataLoader for test data	test_loader
optimizer	Optimizer for updating weights	AdamW with lr=1e-4, weight_decay=1e-4
device	Device to run on (CPU/GPU)	CUDA if available, else CPU
num_epochs	Number of fine-tuning epochs	100

TABLE XII
FINETUNING CONFIGURATION PARAMETERS

3) *Evaluation Process* : The evaluation function provides a comprehensive evaluation of the model's performance. It disables dropout and normalization updates during the evaluation process to ensure consistent inference behaviour. The model extracts the class with the highest probability for each batch, collecting both predictions and ground truth labels for further examination. For HSI classification, while accuracy plays an important role, the cohen kappa score is vital in understanding the reliability of agreement between two raters when classifying the data. This metric takes into account the class imbalance and can therefore prove to be a better metric for understanding the performance of the model.

Parameter	Description	Value Used
model	Model to evaluate	HybridGNNViT instance
dataloader	DataLoader for evaluation data	val_loader or test_loader
device	Device to run on (CPU/GPU)	CUDA if available, else CPU

TABLE XIII
EVALUATION CONFIGURATION PARAMETERS

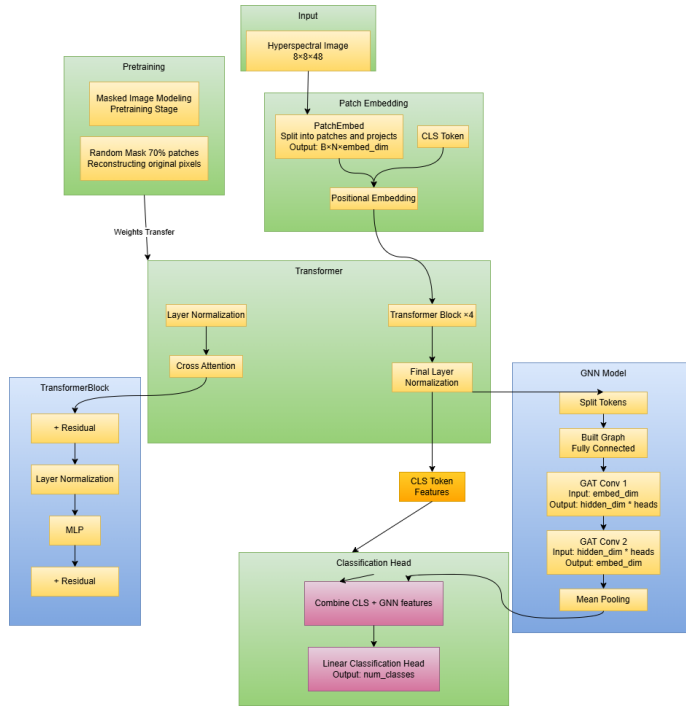


Fig. 1. Model Architecture

CONCLUSION

In this work, we introduced a new hybrid deep learning model Graph Neural Networks and Vision Transformers for hyperspectral image classification, showing a great performance record of 98.52% accuracy on the Houston 2018 data set. Our studies demonstrate that incorporating these complementary architectures can well capture the global contextual information and the local spatial contexts, and the self-supervised masked image modelling pretraining procedure can circumvent the small, labelled data issue in the hyperspectral field. Weighted binary cross-entropy loss function and custom patch embedding methods also improved performance by addressing class imbalance. This work not only pushes the limits of current hyperspectral image analysis but also provides a benchmark for future study in multi-sensor fusion, change detection, and other applications of remote sensing, demonstrating the superiority of a framework that fuses multiple deep learning paradigms to address the challenging issues in hyperspectral remote sensing.

TABLE XIV
LITERATURE REVIEW

S.No	Author(s) & Year	Title	Dataset(s) Used
1	Alexey Dosovitskiy et al. (2021)	An Image is worth 16x16 Words	JFT-300M
2	He, Mingyi (2021)	Spatial-Spectral Transformer for HI Classification	Indian Pines, PaviaU
3	Weili Kong (2023)	A scalable Self-supervised learner for HI classification	GF-5 satellite HI, Indian Pines
4	A. B. Hamida (2018)	Deep Learning Approach for Remote Sensing image classification	Indian Pines, PaviaU, Salinas
5	Xiaofeng Zhao (2025)	A review of HSI classification based on GNN	Indian Pines, PaviaU, Houston 2013
6	Linus Scheibenreif (2023)	Masked vision transformers for HSI classification	EnMAP-DFC, Houston 2018
7	Yao Ding (2024)	GNN for feature extraction & classification of HS remote sensing images	Indian Pines, PaviaU, Houston 2013
8	Chen, Zhang (2023)	Generative SSL with spectral-spatial masking for HS target detection	Indian Pines, PaviaU
9	Nhat-Tuong Do-Tran (2024)	Learning CNN on ViT: a hybrid model	VisDA-C
10	Ashish Vaswani (2017)	Attention is all you need	WMT 2014

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